

Chapter 7: Analytic Trigonometry

Name: _____

Date: _____

Instructions: Answer all questions. Show complete work and justify your steps. Give *exact* values (in terms of π and radicals) unless otherwise specified. Clearly indicate whether angles are in degrees or radians. Each problem begins on a new page.

1. Verify each identity. Work with one side to transform to the other.

(12)

(a) $\frac{2 \tan x}{1 + \tan^2 x} = \sin 2x$

(b) $\tan(\pi - \theta) = -\tan \theta$

- (c) Express $\sin(\theta - \pi)$ using only $\sin \theta$ and $\cos \theta$.

2. Find each exact value.

(10)

(a) $\sin 105^\circ$

(b) $\cos 15^\circ$

(c) $\tan\left(\frac{\pi}{12}\right)$

3. Simplify completely: $\frac{\sin 2x}{\sin x} - \frac{\cos 2x}{\cos x}$. State any domain restrictions. (10)

4. Solve each equation for the stated interval.

(12)

(a) $\sin 3x = \sin x, \quad 0 \leq x < 2\pi$

(b) $\cos 2\theta = 1 - \sin \theta, \quad -\pi \leq \theta \leq \pi$

5. Evaluate exactly: $\tan\left(\arccos\left(-\frac{1}{2}\right) + \arcsin\left(\frac{\sqrt{2}}{2}\right)\right)$. Use principal values for $\arcsin$, $\arccos$. (12)

6. Determine the constants a and b so that (12)

$$\frac{-3 + 4 \cos^2 \theta}{1 - 2 \sin \theta} = a + b \sin \theta.$$

State any identities used and any restrictions on θ where denominators are nonzero.

7. Write $f(x) = 3 \sin x - 4 \cos x$ in the form $R \sin(x - \varphi)$ with $R > 0$ and $-\frac{\pi}{2} < \varphi \leq \frac{\pi}{2}$. (12)
Then find the maximum value of f and the smallest positive x at which the maximum occurs.

8. Show that

$$\frac{\sin x + \sin 3x + \sin 5x}{\cos x + \cos 3x + \cos 5x} = \tan 3x \quad (12)$$

for all x for which both sides are defined.

9. Compute the smallest positive angle x , in degrees, such that (10)

$$\tan 4x = \frac{\cos x - \sin x}{\cos x + \sin x}.$$

10. Compute $(\sin 15^\circ + \sin 75^\circ)^6$. (8)